

Operational Reliability versus Regulatory Severity Emphasis in CFM56-7B Subsystems

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1. Introduction

1.1 Background

Aircraft propulsion system reliability is a central concern for both aviation safety regulators and airline maintenance organizations. Aviation safety frameworks often distinguish between failure frequency and failure consequence when evaluating risk within complex systems. While operational reliability focuses on the rate at which failures occur in service, regulatory intervention tends to prioritize failure modes with severe potential consequences.

Probabilistic Risk Assessment (PRA) frameworks define risk as the product of event probability and consequence severity (Bao, 2018). Within aircraft propulsion systems, failures at the component level can propagate through multiple system layers before producing aircraft-level hazards. One example is the loss of thrust control, which represents a critical link between subsystem malfunction and aircraft safety outcomes (Bao, 2018).

From an operational perspective, propulsion system reliability is also closely tied to airline maintenance economics. Engine maintenance represents approximately 35 to 40 percent of total airline maintenance costs, making propulsion system reliability a major contributor

to operating expenses (Zhang et al., 2020). Unscheduled engine removals further increase maintenance burden and operational disruption.

1.2 Problem Statement

Service Difficulty Reports (SDRs) provide insight into the operational frequency of discrepancies experienced in service. In contrast, Airworthiness Directives (ADs) represent regulatory interventions typically driven by the severity of potential safety consequences.

These two perspectives—operational reliability and regulatory severity emphasis—may not align when examined at the subsystem level. Subsystems that generate frequent operational discrepancies may not receive significant regulatory attention if their consequences are relatively minor, while rare but severe failure modes may dominate regulatory focus.

1.3 Research Question

Do observed CFM56-7B SDR subsystem failure patterns align with regulatory severity emphasis and maintenance economics trends?

1.4 Contribution

This study provides a subsystem-level analysis of 1,419 CFM56-7B Service Difficulty Reports, combining operational reliability data with simplified severity weighting and lifecycle behavior analysis. The results contribute to understanding how operational discrepancy patterns compare with regulatory prioritization within modern turbofan engines.

2. Literature Review

2.1 Probabilistic Risk Assessment Framework

Probabilistic Risk Assessment provides a structured method for evaluating risk within complex engineered systems. Risk is commonly defined as the product of event probability and consequence severity, allowing analysts to compare hazards with differing likelihoods and outcomes (Bao, 2018).

Within aviation propulsion systems, failures may originate at the component level but escalate through system interactions. Loss of thrust control represents a key bridging concept linking component-level failures to aircraft-level consequences (Bao, 2018). PRA literature also distinguishes between individual risk and fleet-level risk exposure, recognizing that rare but catastrophic events may justify regulatory action even when operational frequency is low.

Consequently, aviation regulatory authorities often prioritize high-severity failure modes, particularly those that could result in loss of propulsion or structural failure.

2.2 Operational Reliability and Maintenance Economics

While regulatory frameworks emphasize safety consequences, airline operators often focus on operational reliability and maintenance cost impacts. Aircraft engines account for approximately 35–40 percent of airline maintenance costs, making propulsion system performance a major driver of airline operating economics (Zhang et al., 2020).

Studies of engine lifecycle performance show that approximately 20 percent of engine shop visits occur due to unscheduled removals, which can significantly disrupt airline operations (Zhang et al., 2020). Design upgrades and improved maintenance strategies have been shown to increase time-on-wing performance and shift unscheduled removal distributions toward higher cycle counts.

Thermal degradation indicators such as Exhaust Gas Temperature (EGT) margin are widely used as predictors of long-term engine deterioration and maintenance requirements (Zhang et al., 2020).

2.3 Literature Gap

Existing research tends to emphasize either probabilistic safety modeling or maintenance economics analysis, but rarely compares operational failure frequency patterns with regulatory severity prioritization using empirical field data.

This study addresses that gap by analyzing Service Difficulty Reports for the CFM56-7B turbofan engine to compare operational discrepancy patterns with expected regulatory emphasis.

3. Data Collection

3.1 Service Difficulty Reports

The primary dataset for this study consists of 1,419 Service Difficulty Reports (SDRs) associated with the CFM56-7B turbofan engine. The SDRs were collected directly from the FAA Service Difficulty Reporting System (SDRS) interface. The query used is shown in Figure 3.1:

- Difficulty Date Range: 06/01/2021 – 12/31/2025
- Engine Model: CFM56-7B
- Additional filters for part numbers, aircraft make, and other fields were left broad to capture the full range of relevant discrepancies.

Figure 3.1: Query interface and criteria used to extract SDRs from FAA SDRS.

The following variables were extracted for analysis:

- Failure narrative descriptions – used to classify operational discrepancies into subsystem failure categories.
- PartTotalCycles – used as a measure of engine lifecycle exposure at the time of the reported discrepancy.

It should be noted that SDRs represent reported operational discrepancies rather than exposure-normalized failure probabilities. Therefore, this dataset provides insight into patterns of operational reliability rather than absolute likelihoods of failure.

3.2 Airworthiness Directives

Airworthiness Directives (ADs) were reviewed to identify subsystem areas receiving regulatory attention. AD issuance was used as a proxy indicator of regulatory severity prioritization, since ADs are typically issued when safety concerns with potentially serious consequences are identified.

Because many ADs predate the SDR dataset analyzed in this study, they provide useful context for comparing regulatory focus with operational discrepancy patterns.

4. Methodology

4.1 Failure Family Classification

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Failure narratives from the Service Difficulty Reports (SDRs) were classified into subsystem-level groups using a keyword-based mapping approach. Each narrative was scanned for specific keywords associated with known failure mechanisms, and then assigned to one of the following failure families:

- Fuel and Control – narratives containing terms related to fuel metering, engine control components, actuators, or electrical/hydraulic control systems (e.g., “FUEL,” “THROTTLE,” “EEC,” “FADEC,” “ACTUATOR,” “BLEED,” “HYD”).
- Thermal – narratives containing terms indicative of overheating or temperature anomalies (e.g., “FIRE,” “SMOKE,” “OVERHEAT,” “EGT,” “ANTI ICE”).
- Structural and Rotor – narratives containing terms related to rotor components, structural integrity, or vibration (e.g., “BLADE,” “CRACK,” “VIBRATION,” “SHIM,” “SHROUD,” “LPT,” “HPC”).
- Lubrication and Gearbox – narratives containing terms associated with lubrication, gearbox, or metal wear (e.g., “CHIP,” “OIL,” “AGB,” “METAL”).
- Combustion Instability – narratives containing terms related to flame, stall, surge, or combustion performance (e.g., “STALL,” “FLAME,” “HUNG,” “SURGE”).
- Foreign Object Damage (FOD) – narratives containing terms indicating ingestion of debris or birds (e.g., “FOD,” “BIRD”).

This keyword-based classification enables automated aggregation of individual discrepancies into subsystem-level reliability patterns, supporting quantitative analysis of failure trends and subsystem risk assessment.

4.2 Frequency Analysis

Failure frequency was measured by calculating the count and percentage distribution of SDR events across subsystem categories.

This approach provides a measure of operational reliability burden associated with each subsystem.

4.3 Severity Coding

Each subsystem category was assigned a severity classification based on typical safety consequences associated with those failure types:

- Catastrophic
- Hazardous
- Major
- Minor

These categories were used to approximate the potential safety consequences of subsystem failures.

4.4 Risk Index Construction

A simplified Risk Index was constructed to combine frequency and severity information.

The Risk Index was defined as:

$\text{Risk Index} = (\text{Subsystem Percent of Total Failures}) \times (\text{Average Severity Score})$

This metric provides a simplified probability-weighted comparison of subsystem risk exposure.

4.5 Lifecycle Analysis

Lifecycle behavior was evaluated using PartTotalCycles data. Average cycle counts were calculated for each subsystem category and compared with the fleet average.

This analysis allows identification of:

- Early-life failures
- Wear-out degradation patterns
- Subsystem-specific lifecycle characteristics

4.6 Method Limitations

Several limitations should be noted:

- SDR data is not normalized by flight hours or exposure.
- Narrative classification introduces potential categorization uncertainty.
- The analysis does not include formal event-tree modeling typical of full PRA studies.
- SDR records reflect operational discrepancies rather than validated failure probabilities.

5. Results

5.1 Frequency Distribution

Analysis of the SDR dataset shows that Fuel and Control subsystem discrepancies occur most frequently, accounting for approximately 28% of total failures (404 of 1,419 events), followed by Thermal failures at 26% (367 events). Structural and rotor-related events account for approximately 16% (227 events), while other subsystems contribute smaller shares. This distribution indicates that operational reliability burden is concentrated in Fuel and Control and Thermal subsystems.

5.2 Severity Contrast

Severity weighting reveals a different pattern. Rotor-related failures, while less frequent, exhibit a higher proportion of catastrophic severity classifications. In contrast, Fuel and Control failures tend to produce moderate operational disruptions rather than catastrophic hazards.

This contrast highlights a divergence between operational reliability burden and safety consequence severity.

5.3 Risk Index Ranking

When combining frequency and severity using the Risk Index metric, the Thermal subsystem ranks highest with a composite score of 828, exceeding Fuel and Control (786) despite its slightly lower frequency. This indicates that while Fuel and Control dominates in operational occurrence, Thermal failures represent greater overall risk exposure when severity is considered. Combustion Instability follows with a Risk Index of 388, while Structural and Rotor-related failures rank lower at 281, reflecting their lower frequency despite potential high consequences.

5.4 Lifecycle Clustering

Lifecycle analysis shows distinct subsystem behavior patterns. Fuel and Control failures occur at an average of approximately 11,230 cycles, below the fleet average of 12,506 cycles, suggesting early-life reliability sensitivity. In contrast, Thermal failures occur at

higher average cycle counts of approximately 14,502 cycles, consistent with wear-out degradation processes such as thermal stress accumulation and EGT margin deterioration. Structural and Rotor-related failures also occur at elevated cycle counts (~13,563 cycles), indicating a tendency toward later-life failure behavior.

5.5 Survival Analysis of Subsystem Failures

To further evaluate lifecycle behavior across subsystem categories, a survival analysis approach was applied using PartTotalCycles as a proxy for lifecycle exposure. Kaplan–Meier survival curves were constructed for each failure family to visualize the distribution of failure occurrence across engine operating cycles. These patterns are consistent with observed average lifecycle metrics, where Fuel and Control failures occur at approximately 11,230 cycles compared to 14,502 cycles for Thermal failures, reinforcing the distinction between early-life discrepancies and degradation-driven failure mechanisms.

Figure 1 shows the survival probability of each subsystem as a function of total accumulated cycles. In this context, survival probability represents the likelihood that a component has not yet experienced a reported discrepancy of a given failure type at a specific cycle count.

The results reveal distinct lifecycle patterns across subsystems. Fuel and Control failures exhibit a relatively steep early decline in survival probability, indicating a higher concentration of discrepancies occurring at lower cycle counts. This suggests early-life reliability sensitivity, potentially associated with control system tuning, component tolerances, or integration effects.

In contrast, Thermal-related failures demonstrate a more gradual decline initially, followed by a sharper decrease at higher cycle counts. This pattern is consistent with wear-out mechanisms driven by thermal degradation processes, such as Exhaust Gas Temperature (EGT) margin deterioration and material fatigue.

Structural and Rotor-related failures appear less frequently in early lifecycle stages but extend across a broader range of higher cycle counts. This behavior supports their characterization as lower-frequency but higher-consequence events, aligning with regulatory emphasis on catastrophic failure modes.

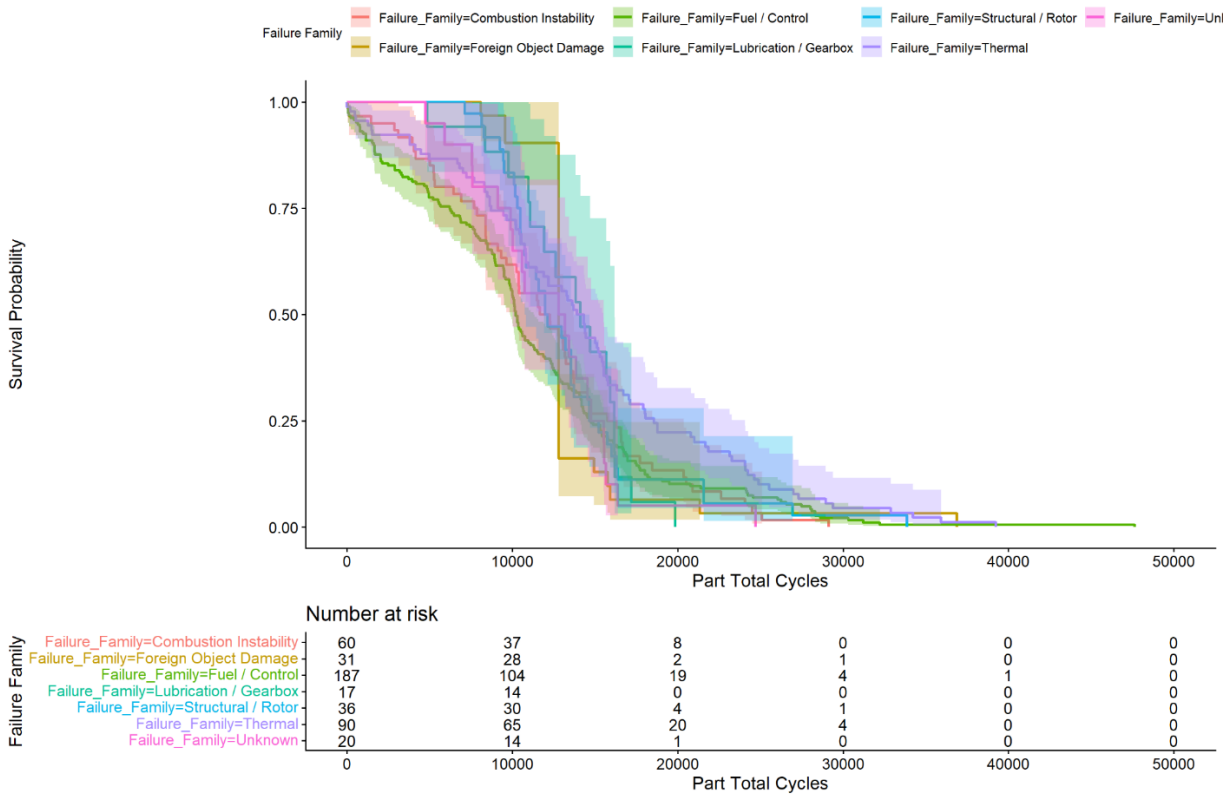
Foreign Object Damage (FOD) events display less consistent lifecycle dependence, reflecting their exogenous nature rather than intrinsic degradation processes.

It should be noted that survival estimates are derived from Service Difficulty Report data and therefore reflect observed discrepancy timing rather than exposure-normalized failure probabilities. As such, the analysis is best interpreted as a comparative lifecycle distribution of failure occurrences rather than a true reliability function.

Overall, the survival analysis reinforces the distinction between early-life operational discrepancies and long-term degradation-driven failures, providing additional evidence that subsystem reliability behavior varies significantly across the engine lifecycle. These patterns suggest opportunities for differentiated maintenance strategies, including early-life reliability improvements for control systems and condition-based monitoring approaches for thermal degradation.

Figure 1.

Kaplan–Meier survival curves for CFM56-7B subsystem failure families based on PartTotalCycles.



6. Conclusion

This study analyzed 1,419 CFM56-7B Service Difficulty Reports to compare operational reliability patterns with regulatory severity emphasis.

Results indicate that operational discrepancy frequency is dominated by Fuel and Control and Thermal subsystems, which represent the largest share of reported failures. In contrast, regulatory attention remains heavily focused on low-frequency rotor failures due to their potential catastrophic consequences.

Lifecycle analysis further shows that subsystem failures exhibit distinct degradation patterns, with control system discrepancies appearing earlier in engine life and thermal degradation becoming more prominent at higher cycle counts.

These findings suggest that current regulatory emphasis may not fully capture the operational reliability burden experienced by airlines, particularly in high-frequency

subsystems such as Fuel and Control. Conversely, low-frequency rotor failures continue to dominate regulatory attention due to their catastrophic potential. This divergence highlights the need for a more integrated risk framework that simultaneously accounts for lifecycle reliability, maintenance cost impact, and safety consequence severity.

From an operational perspective, high-frequency subsystems such as Fuel and Control—accounting for approximately 28% of all reported failures—may drive a disproportionate share of maintenance burden despite receiving comparatively less regulatory emphasis.

References

Bao, Y. (2018). Probabilistic risk assessment of aircraft propulsion systems and loss-of-thrust-control events.

Zhang, X., Li, Y., & Chen, H. (2020). Lifecycle maintenance economics and performance degradation in commercial turbofan engines. *Energies*, 13(9).